

Worksheet

For each problem construct an integral which solves the given problem.

Area

1. Find the area of the bounded region bounded between the curves
 $y = x^2 - 2x$, $y = 6 - x$
2. Find the area of the region bounded between the curves $y = x^2 - 2x$, $y = x$
between $x = 0$ and $x = 4$.
3. Find the area of the bounded region bounded by the curves
 $y = 2x + 5$, $y = 13 - 3x$, $y = x^2 - 2x - 7$
4. Find the area of the bounded region in the first quadrant bounded between the curve
 $x = -2\frac{y-9}{y+3}$ and the x -axis and the y -axis.
5. Find the area of the bounded region in the first quadrant bounded between the curves
 $y = \sqrt{x}$, $y = x - 6$, x -axis using vertical strips.
6. Find the area of the bounded region in the first quadrant bounded between the curves
 $y = \sqrt{x}$, $y = x - 6$, x -axis using horizontal strips.
7. Find the area of the bounded region bounded between the curves
 $y = x - 1$, $y^2 = 2x + 6$ using vertical strips.
8. Find the area of the bounded region bounded between the curves
 $y = x - 1$, $y^2 = 2x + 6$ using horizontal strips.